

Lecture 1: Complex Variables With Applications

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Definition 1 (differentiable closed loop). γ is called a differentiable closed loop in $\mathbb{C}$ if γ can be parameterized by

$$\{z = f_1(s) + f_2(s)i : s \in (a, b]\}$$

with f_1 and f_2 satisfying

- (1). For all $s \in (a, b]$, f_1 and f_2 are two differentiable real-valued functions of the variable s ;
- (2). For any s_1, s_2 in $(a, b]$ with $s_1 \neq s_2$, it holds $f_1(s_1) + f_2(s_1)i \neq f_1(s_2) + f_2(s_2)i$
- (3). The following two limits hold

$$\lim_{s \rightarrow a+} f_1(s) = f_1(b), \quad \lim_{s \rightarrow a+} f_2(s) = f_2(b)$$

Example 2 (discontinuous at origin).

$$f(z) := \begin{cases} z/\bar{z}, & \text{if } z \neq 0 \\ 1, & \text{if } z = 0 \end{cases}$$

For $z = x + iy \neq 0$, the real part of this f is read as

$$\operatorname{Re}(f) = \frac{x^2 - y^2}{x^2 + y^2}$$

If we approach the origin along $(x, 0)$, then $\operatorname{Re}(f)(x) = 1$ for all x . If we approach the origin along $(0, y)$, then $\operatorname{Re}(f)(iy) = -1$ for all y . Clearly $\operatorname{Re}(f)$ has no limit while (x, y) approaches to 0. Therefore no matter what value we assign for f at the origin, f can never be continuous at 0.

[BS10]
[CRT06]

References

- [BS10] Z. Bai and J. Silverstein, *Spectral Analysis of Large Dimensional Random Matrices*, Springer, 2010.
- [CRT06] E. Candes, J. K. Romberg, and T. Tao, *Stable signal recovery from incomplete and inaccurate measurements*, Communications on Pure and Applied Mathematics **59** (2006), 1207–1223.